

Finite element analysis (FEA) of a flexible bead chain requires thousands of force and integration computations per simulated second. We train a graph neural network (GNN) to predict the per-timestep physics of a 1D bead chain with tapered mass, using randomly sampled single-step training data instead of recorded simulations. An initial shared MLP achieved accurate predictions on a fixed 16-bead chain but could not generalize to other lengths. We restructure the surrogate as a true message-passing GNN with separate node/edge encoders, learned messages per rod, and ghost masking for endpoints. Because all weights are shared, one trained model handles chains of any length.

A whip is modeled as a 1D chain of  $N$  tapered mass nodes connected by  $N - 1$  spring rods, anchored at one end. Each free bead experiences gravity, Hookean spring forces, rod-axis damping, and velocity drag. Time integration uses symplectic Euler:

$$\mathbf{v}_i^{n+1} = \mathbf{v}_i^n + \Delta t \mathbf{a}(\mathbf{x}^n) \quad (1)$$

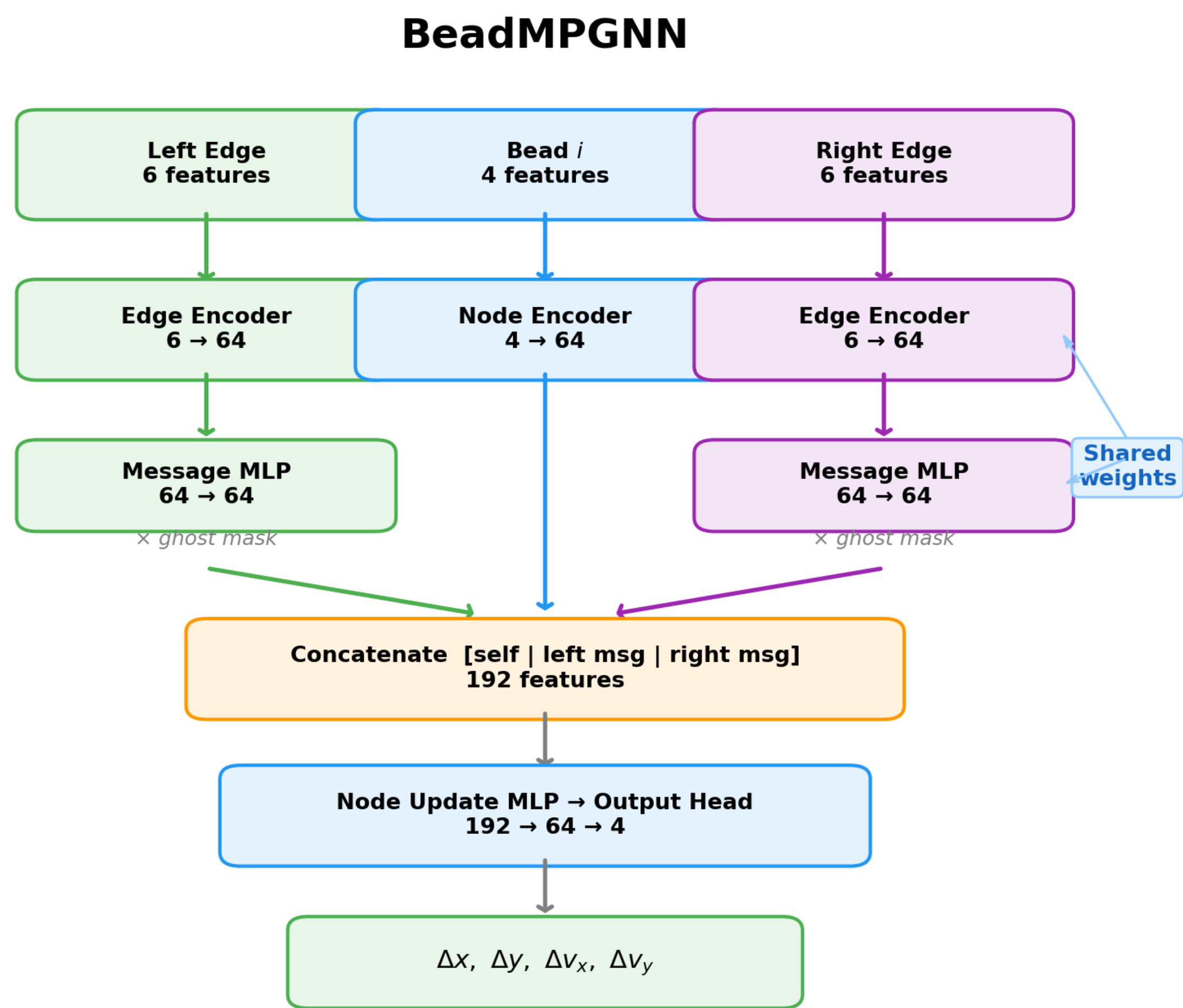
$$\mathbf{x}_i^{n+1} = \mathbf{x}_i^n + \Delta t \mathbf{v}_i^{n+1} \quad (2)$$

The force on bead  $i$  depends only on its state and those of its immediate neighbors. This locality is why we use a 1-hop graph architecture.

- **Node encoder** ( $4 \rightarrow 64$ ):  $[v_x, v_y, m, \text{fixed}]$
- **Edge encoder** ( $6 \rightarrow 64$ ): shared left/right, relative features
- **Message MLP** ( $64 \rightarrow 64$ ): per-edge learned message
- **Ghost mask**: zero messages from absent neighbors
- **Node update MLP** ( $192 \rightarrow 64$ ): fuses self + messages
- **Output head** ( $64 \rightarrow 4$ ): change in state

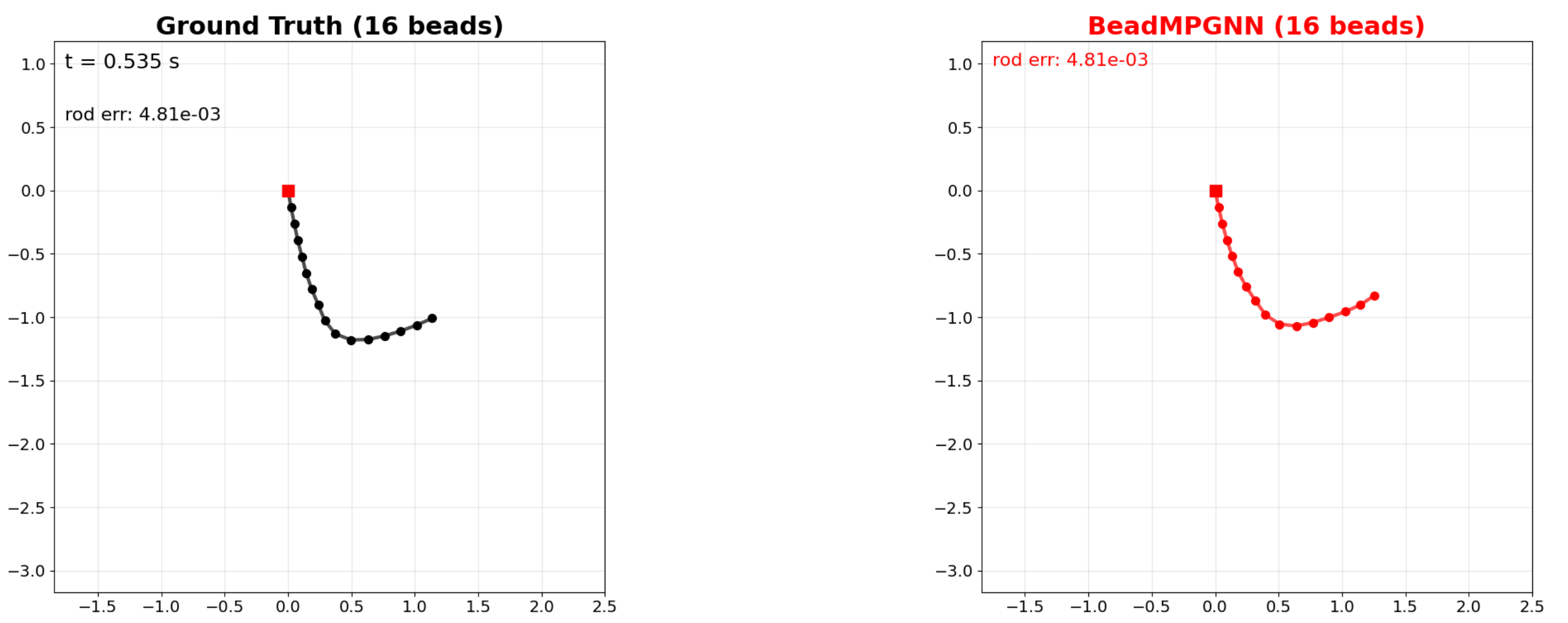
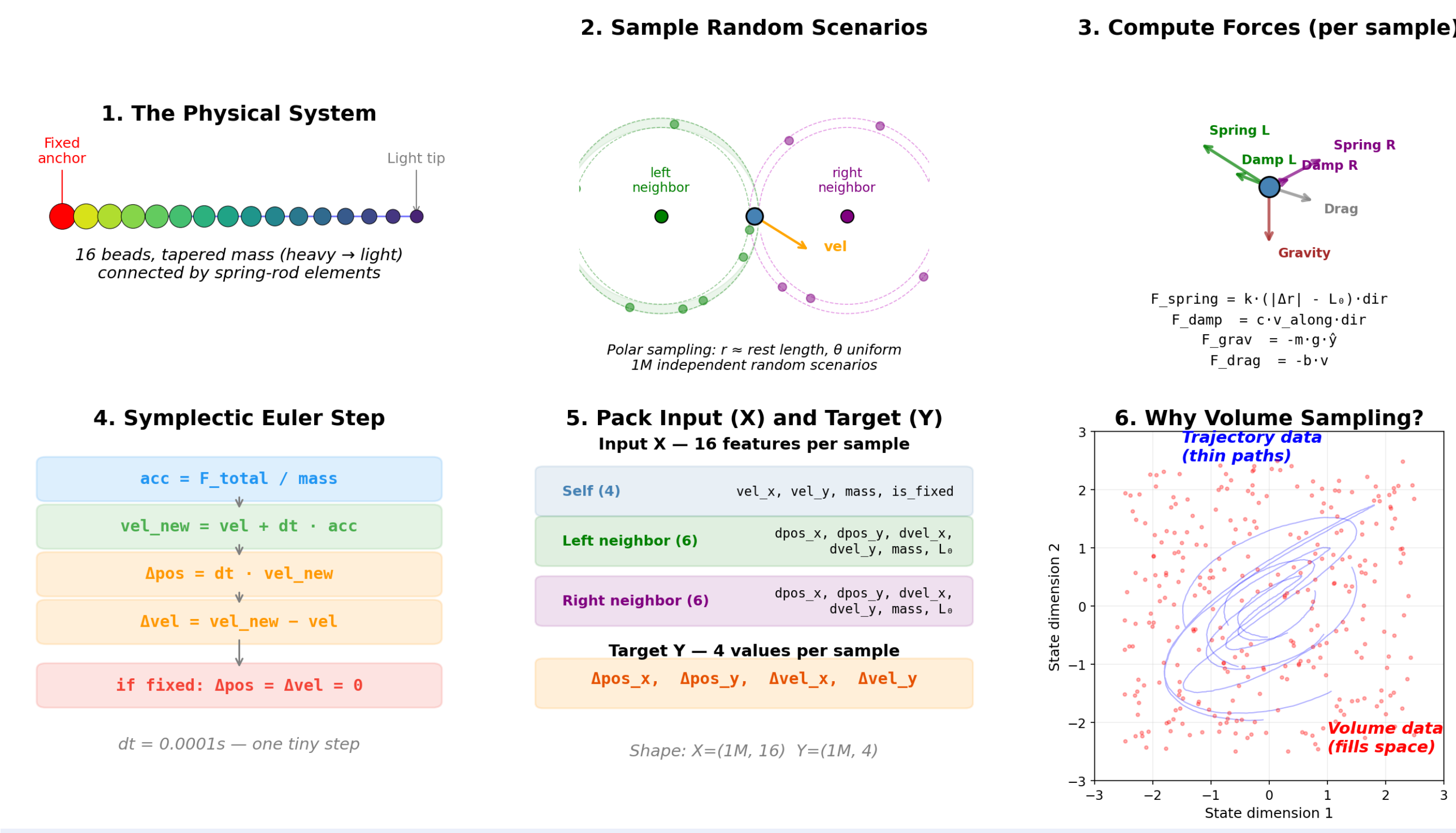
80,324 parameters, shared across all beads, independent of chain length.

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Rather than recording trajectories, we sample random single-step scenarios uniformly across the reachable state space. Each sample is one bead at one instant plus the exact physics result one timestep later. Volume sampling gives the model coverage across the full state space, not just along specific trajectories.

generate\_volume\_data.py — Pipeline Visualization



Ground truth (left, black) vs BeadMPGNN (right, red) on a 16-bead chain at  $t = 0.535$  s.

